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clear all;
clc;

%简单的测试数据集
X = [3,3;4,3;1,1];
x_1 = X(:,1);
x_2 = X(:,2);
Y = [1,1,-1];
%标签 新数据用来验证SVM
Z=[1,0;2,2;4,4];

m = size(X,1);
for i = 1:m
    X(i,:) = X(i, :)*Y(i);
end

%% 对偶问题，用二次规划来求解
H = X*X';

f = [-1;-1;-1];
A = Y;
b = 0;
lb = zeros(3,1);
% 调用二次规划的函数
[alpha,fval,exitflag,output,lambda] = quadprog(H,f,[],[],A,b,lb);
%fmincon(H,X0,A,B)
% 求原问题的解
n = size(alpha,1);
w = alpha' * X;

for i = 1:n
    if alpha(i) > 0
        b = Y(i)-w*X(i,:);
        break;
    end
end

% 求出分离超平面
y_1 = [0,4];
for i = 1:2
    y_2(i) = (-b-w(1)*y_1(i))/w(2);
end
% %斜率和截距
% slope = -w(1)/w(2);
% intercept = -b/w(2);
% % 定义x坐标范围
% x = 0:4;
% % 计算直线的y坐标
% y = slope * x + intercept;
% % 绘制直线
% plot(x, y);

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```
hold on
plot(y_1,y_2,'r');
for i = 1:3
    if Y(i) == 1
        plot(x_1(i,:),x_2(i,:),'+r');
    elseif Y(i) == -1
        plot(x_1(i,:),x_2(i,:),'og');

    end
    plot(Z(i,1),Z(i,2),'+b');
end
```

hold off

```
%axis([0,7,0,7])
hold off
```